

Rent Impromptu

Team 6: Lana Barakat, Charles Doan, Joyce Dong,
Abdallah Khan, Aidan Valva, Vinh Vu

Objective of Project

Allow users to share/rent less frequently used items.

- Save money
- Save storage space

Web-app-based peer-to-peer rental marketplace

Safe, user-friendly space to rent or rent out items and generate extra income on the side



Project Timeline

- Start Date: December 1, 2025
- End Date: June 1, 2026
- Duration ~6 months (24 weeks)

Justification:

A six-month schedule provides ample time to complete the full life cycle of the project– which consists of planning, development, integration, testing, and deployment–without rushing critical tasks. By structuring the work into incremental sprints, each phase delivers tangible progress and reduces risks, allowing early feedback and smoother integration of features.

This timeline also ensures that testing, performance tuning, and final refinements are handled thoroughly rather than compressed at the end. The built-in buffer adds flexibility for unexpected issues/events, which helps maintain quality while still meeting the target delivery date.



Cost Estimation

PostgreSQL: \$0.00

Labor: 6 workers \$20/hr for 24 Weeks (40 hours/week) = \$115,200

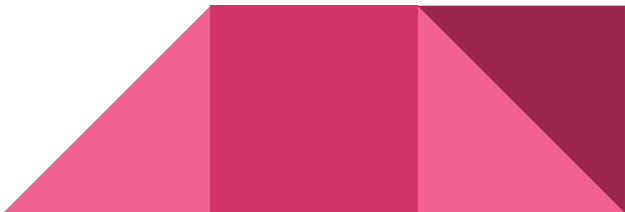
Hardware: Macbook air \$750/per person = \$4500

Servers: AWS EC2 \$75/month = \$450

Database: AWS RDS \$50/month= \$300

Storage: AWS S3 100GB \$25/month= \$150

Total = \$120,600



Functional Requirements Overview

- User Account Management
- Item Listing Management
- Item Search and Filtering
- Renting and Reservations
- Payments and Deposits
- Reviews and Ratings
- Customer Support



Non-Functional Requirements

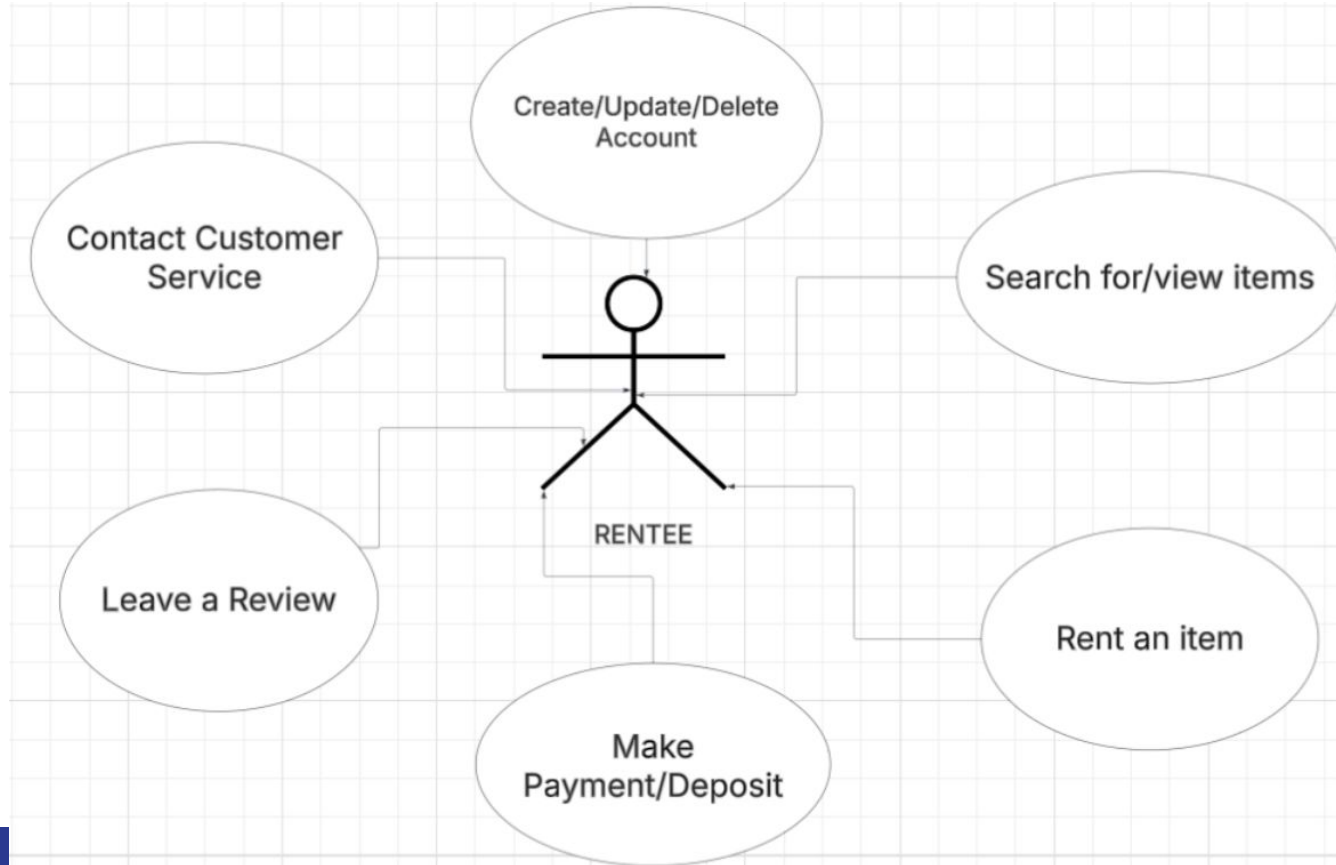
Technical Requirements

- Performance
- Usability
- Scalability
- Maintainability
- Compatibility
- Space
- Dependability
- Security

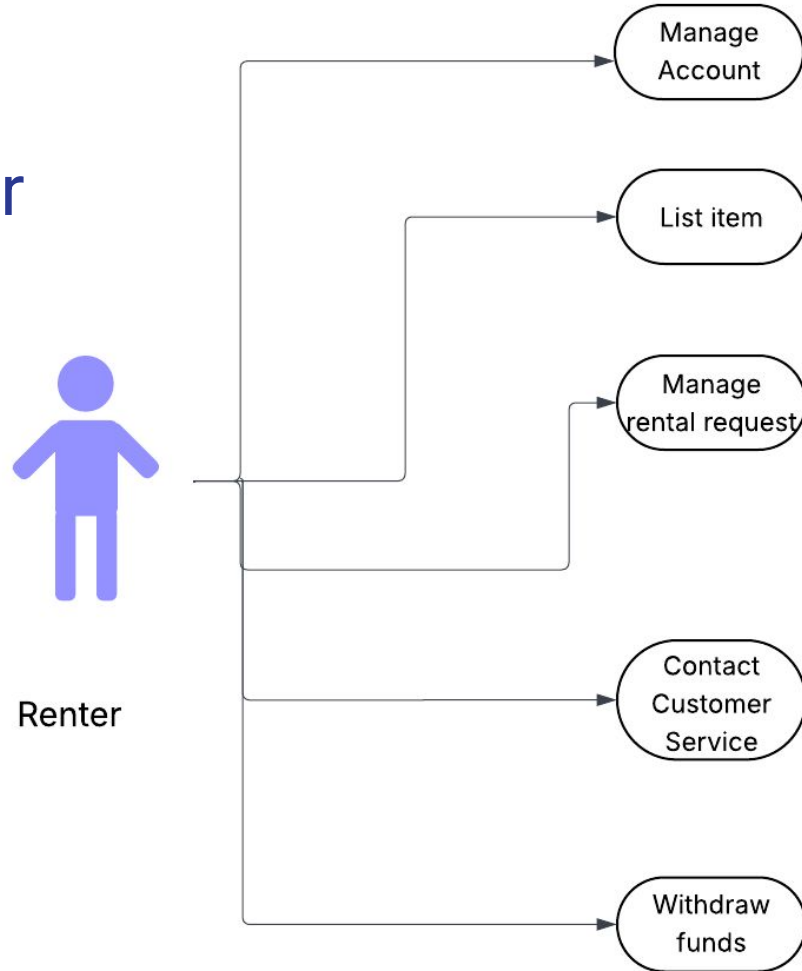
Administrative + Regulatory

- Space
- Operational
- Developmental
- Ethical
- Accounting
- Safety
- Regulatory

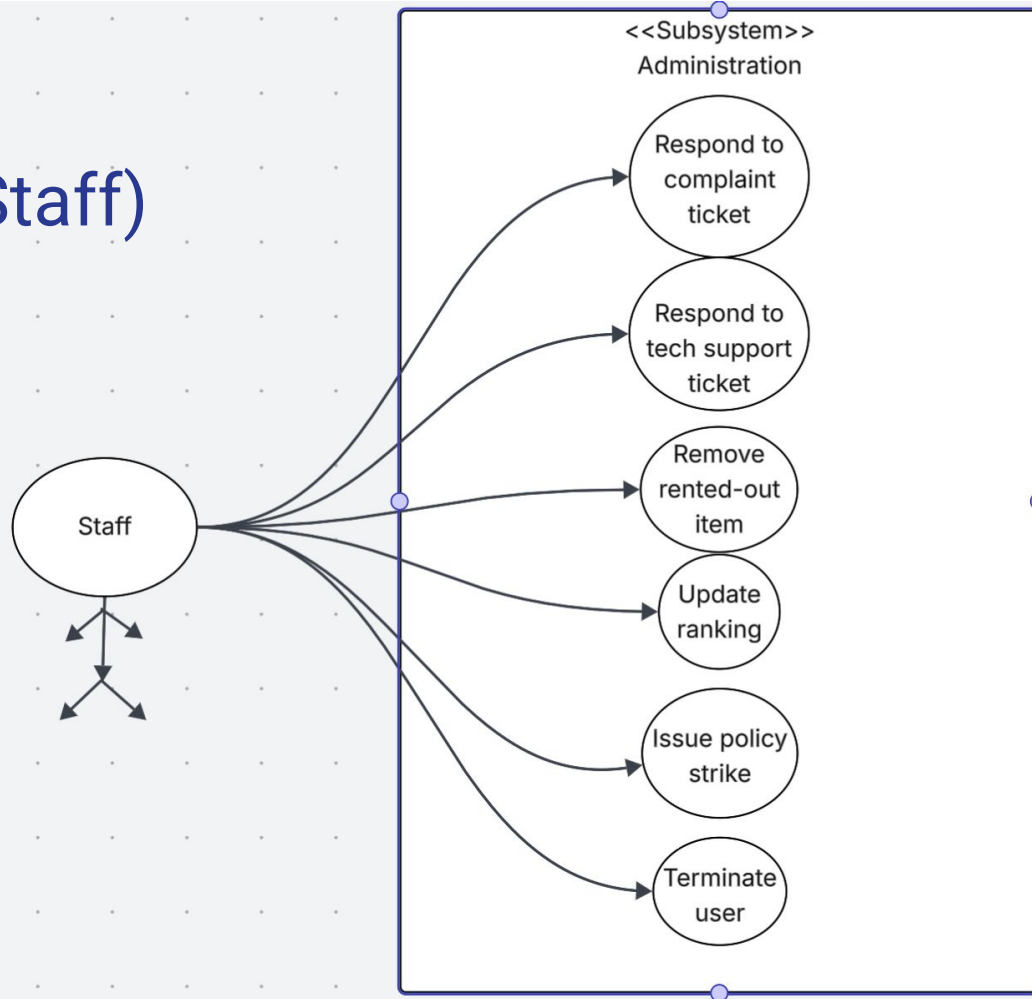
Use Case Diagram (Rentee)



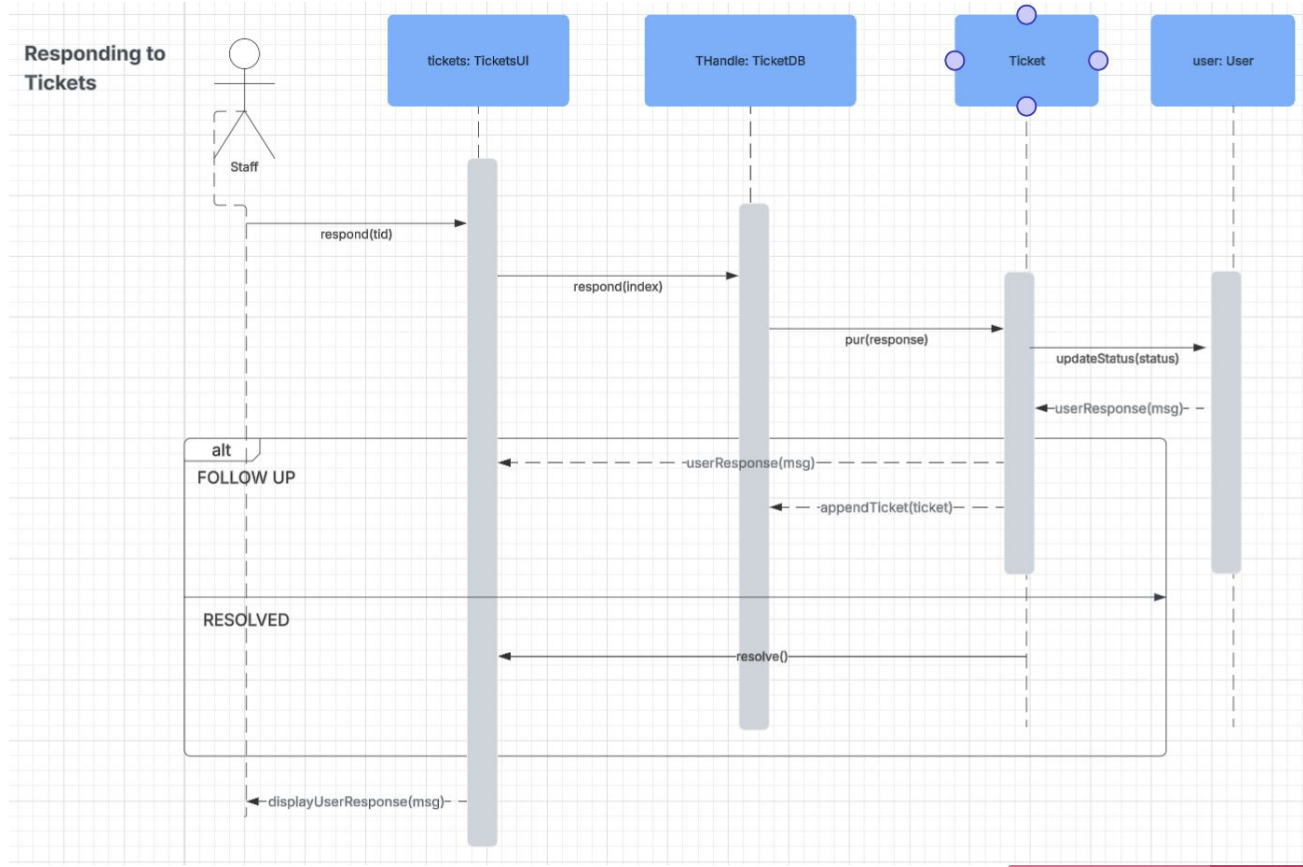
Use Case Diagram (Renter Diagram)



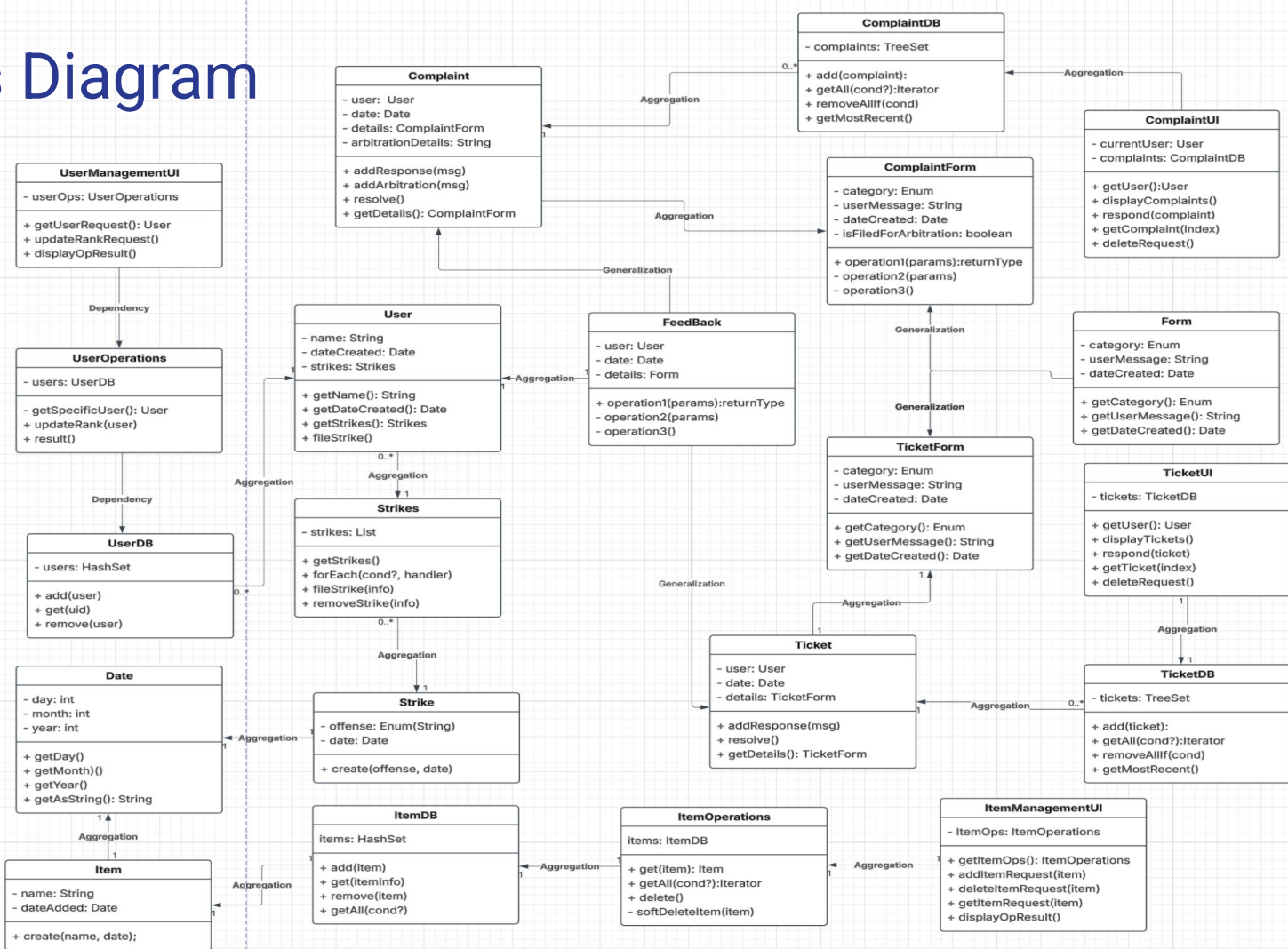
Use Case Diagram (Staff)



Sequence Diagram



Class Diagram



Architectural Design

Client-Server

